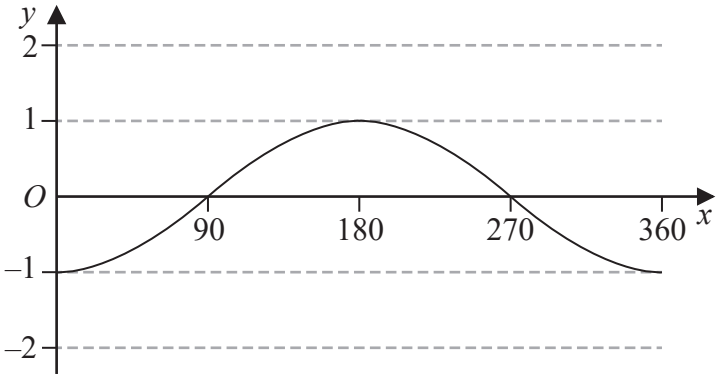
22 Here is a sketch of the graph of a trigonometric function for $0 \leqslant x \leqslant 360$



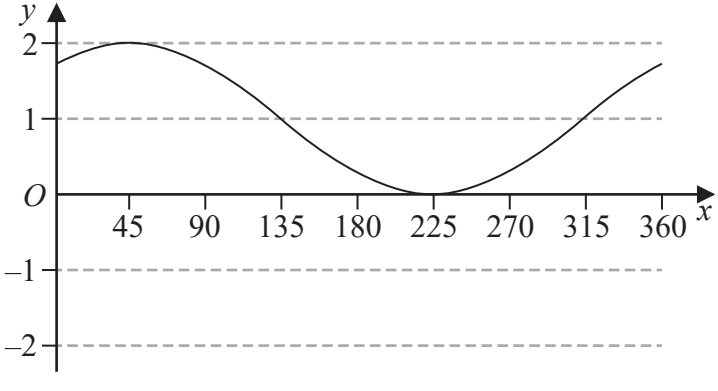
(a) Write down a possible equation of the graph.

$y = -\cos x°$

1 mark
Negative cosine; period 360°.

(1)

Here is a sketch of the graph of $y = \cos(x - p)° + r$ for $0 \leqslant x \leqslant 360$



(b) Find the value of p and the value of r.

$p = 45°$ (right shift)

$r = 1$ (up 1)

1 mark
Maximum at 45°.

1 mark
Midline y = 1.

$p =$ .....

$r =$ .....

(2)

(Total for Question 22 is 3 marks)